

Zhaoyi Wang

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SUMMARY

AI-driven remote sensing and 3D spatial analysis for environmental and ecosystem monitoring, with a focus on LiDAR point clouds, photogrammetry, and multi-temporal change detection applied to natural systems, including terrain dynamics, vegetation structure, and forest-relevant environments.

EDUCATION

ETH Zurich

PhD in Geomatics, Institute of Geodesy and Photogrammetry

- Thesis: "Cross-modal fusion of point clouds and RGB images for landslide monitoring"

University of Science and Technology of China

MSc in Precision Engineering Geodesy

Changsha University of Science and Technology

BSc in Surveying and Mapping Engineering

Zurich, Switzerland

Oct. 2021 - Nov. 2025

Hefei, China

Sep. 2018 - Jun. 2021

Changsha, China

Sep. 2014 - Jun. 2018

PROFESSIONAL EXPERIENCE

Geosensors and Engineering Geodesy, ETH Zurich

Research Assistant

Zurich, Switzerland

Oct. 2021 - Dec. 2025

- Developed AI-enabled geospatial pipelines for environmental monitoring, projecting TLS point clouds into panoramic images and detecting artificial targets using customized deep learning detectors (Faster R-CNN, YOLO-X, YOLO-v7)
- Proposed cross-modal feature fusion methods and developed end-to-end geospatial pipelines for robust point cloud registration and multi-epoch 3D displacement estimation in complex natural environments, integrating TLS point clouds and RGB images
- Implemented GPU-accelerated algorithms (PyTorch + CUDA) for large-scale environmental 3D data processing
- Designed automated workflows for preprocessing, depth map consistency, visibility tracking, and evaluation across multi-epoch UAV datasets over natural terrain and environmental monitoring scenarios
- Application focus: environmental monitoring, natural terrain and vegetation dynamics, and ecosystem-relevant 3D change analysis

3DOM, Fondazione Bruno Kessler (FBK)

Visiting Researcher

Trento, Italy

Sep. 2025 - Nov. 2025

- Developed and benchmarked multi-scene, multi-sensor 3D change detection methods, focusing on generalization across diverse spatial scales, point densities, and environmental conditions (e.g., terrain complexity, vegetation cover)
- Contributed to automated pipelines for multi-epoch UAV RGB reconstruction and 3D surface change detection for UAV-based landslide monitoring

Dilusense Co., Ltd.

Computer Vision Algorithm Intern

Hefei, China

Mar. 2021 - Jun. 2021

- Developed algorithms for RGB-D camera calibration and depth estimation in structured light systems

ADDITIONAL EXPERIENCE

- Teaching Assistant for Geomonitoring and Geosensors (2022), Geospatial Data Acquisition (2023, 2024)
- Volunteer Interpreter, University of Science and Technology of China (Sep. 2018 - Jun. 2021)
- International Geoinformatics Summer School, Wuhan University (Aug. 2020)
- Exchange Program, Hong Kong Polytechnic University (Jul. 2017)

SELECTED ACTIVITIES AND ACHIEVEMENTS

- Best Presentation Award, ISPRS Geospatial Week – Laser Scanning Workshop (2023)
- National Scholarship, University of Science and Technology of China, Top 2% (2020)
- Top Grade Scholarship, Changsha University of Science and Technology, Top 1% (4×), 2015–2018
- Reviewer: ISPRS Journal of Photogrammetry and Remote Sensing, *International Journal of Applied Earth Observation and Geoinformation*, *Measurement*, *Computational Urban Science*, and *The European Journal on Artificial Intelligence*

CORE SKILLS

- Application Domains: Environmental monitoring, ecosystem and vegetation analysis, geohazard and terrain dynamics
- Remote Sensing & 3D Spatial Data: LiDAR (TLS/UAV), Photogrammetry, SfM/MVS, multi-temporal 3D change detection, geohazard monitoring, object detection, geometric feature extraction
- AI & Programming: Python, C++, Shell scripting, MATLAB, CUDA, Git, LaTeX, Linux, PyTorch, Open3D, OpenCV, COLMAP
- Software: Leica Cyclone, RIEGL RiSCAN, FARO SCENE, Agisoft Metashape, CloudCompare, Unity
- Languages: Mandarin (Native), English (Professional)
- Extracurricular: Half-marathon running, skiing, hiking, table tennis

SELECTED PUBLICATIONS AND SUBMISSIONS

Wang, Z.Y., Trybala, P., Wieser, A., & Remondino, F. (2026). MultiChange3D: A multi-scene, multi-sensor dataset for benchmarking 3D geometric change detection. Under review.

Wang, Z.Y., Butt, J.A., Huang, S.Y., Medić, T., & Wieser, A. (2026). Dense 3D displacement estimation for landslide monitoring via fusion of TLS point clouds and embedded RGB images. *International Journal of Applied Earth Observation and Geoinformation*.

Wang, Z.Y., Butt, J.A., Huang, S.Y., Meyer, N., Medić, T., & Wieser, A. (2025). An approach for RGB-guided dense 3D displacement estimation in TLS-based geomonitoring. *ISPRS Annals, ISPRS Geospatial Week 2025*. (Oral presentation)

Wang, Z.Y., Huang, S.Y., Butt, J.A., Cai, Y.Z., Varga, M., & Wieser, A. (2025). Cross-modal feature fusion for robust point cloud registration with ambiguous geometry. *ISPRS Journal of Photogrammetry and Remote Sensing*.

Wang, Z.Y., Varga, M., Medić, T., & Wieser, A. (2023). Assessing the alignment between geometry and colors in TLS colored point clouds. *ISPRS Annals, ISPRS Geospatial Week 2023*. (Oral presentation)